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JEE MAIN · QUESTION PAPER

JEE Main 2024 (Online) 29th January Morning Shift

JEE Main Question Paper with Answer Key

29 Jan 2024	90	English
HELD ON	QUESTIONS	LANGUAGE

INSIDE THIS PDF

Chemistry	30 questions
Mathematics	30 questions
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Answer key	at the end

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Chemistry 30 questions

Q1. MCQ (Single Correct Answer)

Identify the incorrect pair from the following :

(A) Fluoroapatite $-3Ca_3(PO_4)_2 \cdot CaF_2$

(B) Carnallite $-KCl \cdot MgCl_2 \cdot 6H_2O$

(C) Cryolite $-Na_3AlF_6$

(D) Fluorspar $-BF_3$

Open solution

Q2. MCQ (Single Correct Answer)

Given below are two statements : one is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$:

Assertion A : The first ionisation enthalpy decreases across a period.

Reason $\mathbf{R}$: The increasing nuclear charge outweighs the shielding across the period.

In the light of the above statements, choose the most appropriate from the options given below :

(A) $\mathbf{A}$ is false but $\mathbf{R}$ is true

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

(C) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(D) $\mathbf{A}$ is true but $\mathbf{R}$ is false

Open solution

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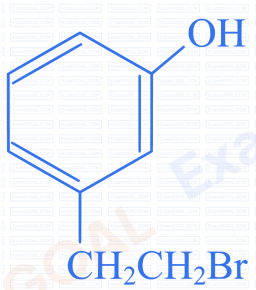
Q3. MCQ (Single Correct Answer)

Open solution

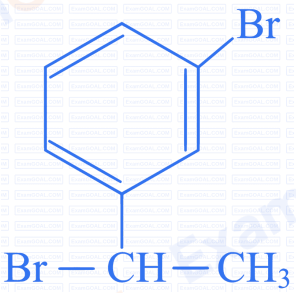
The major product(P) in the following reaction is



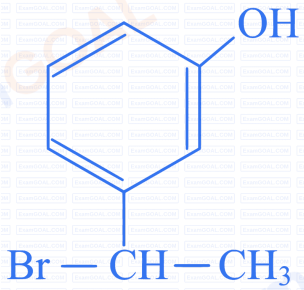
(A)



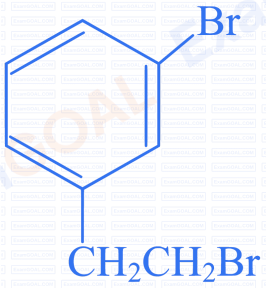
(B)



(C)



(D)

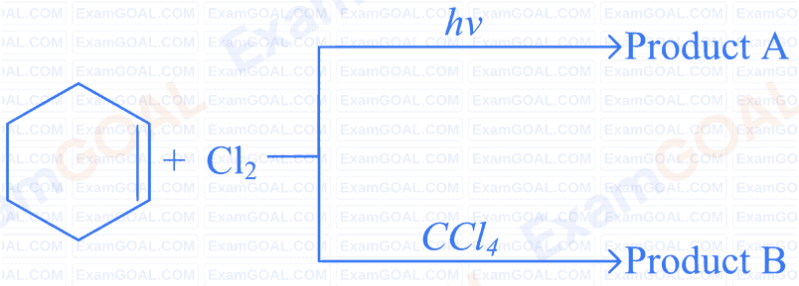


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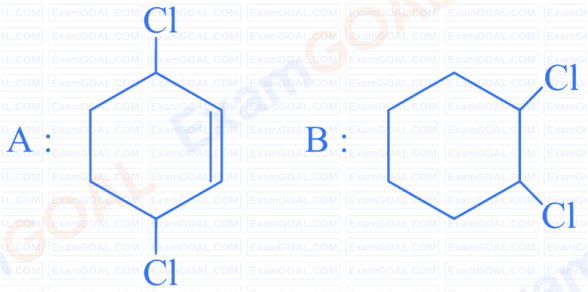
Q4. MCQ (Single Correct Answer)

Open solution

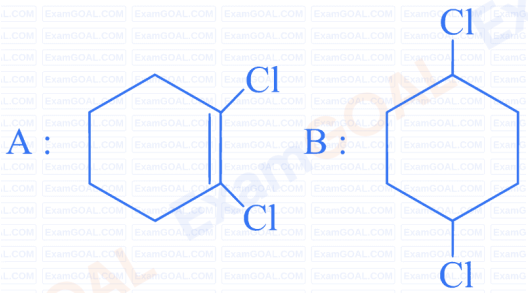
Identify product A and product B :



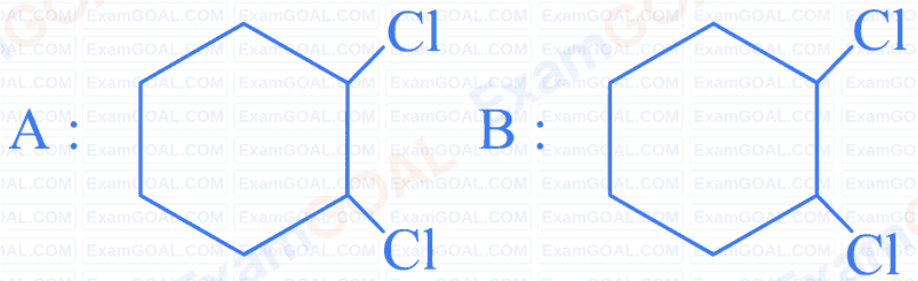
(A)



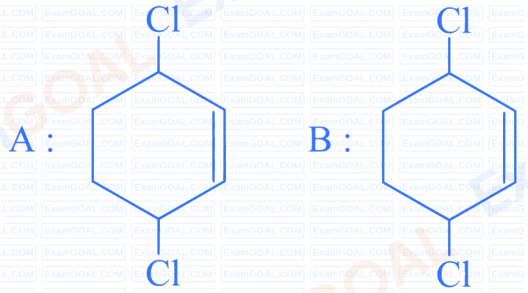
(B)



(C)



(D)

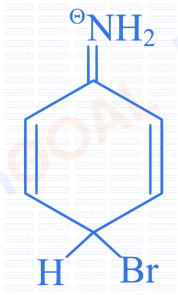


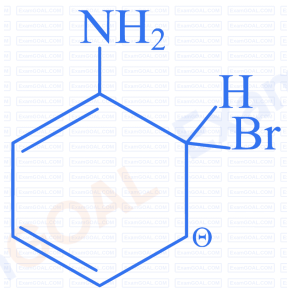
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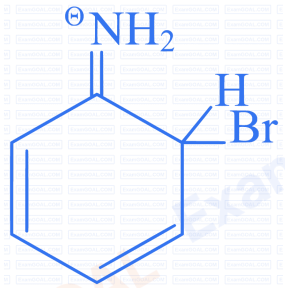
Q5. MCQ (Single Correct Answer)

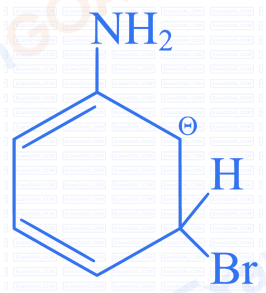
Open solution

The arenium ion which is not involved in the bromination of Aniline is _____.

(A) 

(B) 

(C) 

(D) 

Q6. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : The electronegativity of group 14 elements from Si to Pb, gradually decreases.

Statement II : Group 14 contains non-metallic, metallic, as well as metalloid elements.

In the light of the above statements, choose the **most appropriate** from the options given below :

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are true

(C) Both Statement I and Statement II are false

(D) Statement I is true but Statement II is false

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Q7. MCQ (Single Correct Answer)

Chlorine undergoes disproportionation in alkaline medium as shown below :

$$a\mathrm{Cl}_{2(g)} + b\mathrm{OH}^-_{(aq)} \rightarrow c\mathrm{ClO}^-_{(aq)} + d\mathrm{Cl}^-_{(aq)} + e\mathrm{H_2O}_{(l)}$$

The values of a, b, c and d in a balanced redox reaction are respectively :

(A) 3, 4, 4 and 2 (B) 1, 2, 1 and 1 (C) 2, 4, 1 and 3 (D) 2, 2, 1 and 3

Open solution

Q8. MCQ (Single Correct Answer)

In chromyl chloride test for confirmation of Cl^- ion, a yellow solution is obtained. Acidification of the solution and addition of amyl alcohol and $10\%\ \mathrm{H_2O_2}$ turns organic layer blue indicating formation of chromium pentoxide. The oxidation state of chromium in that is

(A) +6 (B) +5 (C) +3 (D) +10

Open solution

Q9. MCQ (Single Correct Answer)

Type of amino acids obtained by hydrolysis of proteins is :

(A) γ (B) β (C) α (D) δ

Open solution

Q10. MCQ (Single Correct Answer)

The correct set of four quantum numbers for the valence electron of rubidium atom $(\mathrm{Z} = 37)$ is :

(A) $5, 1, 1, +\frac{1}{2}$ (B) $5, 0, 0, +\frac{1}{2}$ (C) $5, 0, 1, +\frac{1}{2}$ (D) $5, 1, 0, +\frac{1}{2}$

Open solution

Q11. MCQ (Single Correct Answer)

The difference in energy between the actual structure and the lowest energy resonance structure for a given compound is

(A) electromeric energy (B) resonance energy

(C) ionization energy (D) hyperconjugation energy

Open solution

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Q12. MCQ (Single Correct Answer)

Open solution

Match List I with List II

	List - I (Substances)		List - II (Element Present)
(A)	Ziegler catalyst	(I)	Rhodium
(B)	Blood Pigment	(II)	Cobalt
(C)	Wilkinson catalyst	(III)	Iron
(D)	Vitamin B_{12}	(IV)	Titanium

Choose the correct answer from the options given below:

(A) A-IV, B-III, C-I, D-II

(B) A-II, B-IV, C-I, D-III

(C) A-III, B-II, C-IV, D-I

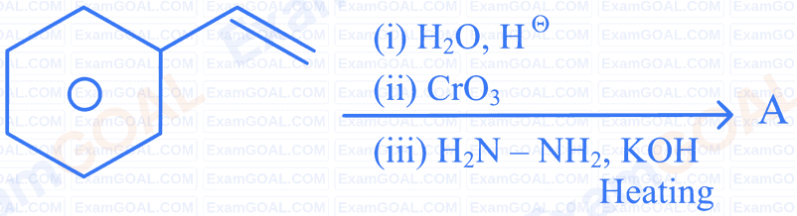
(D) A-II, B-III, C-IV, D-I

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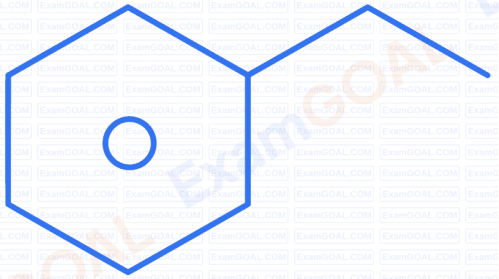
Q13. MCQ (Single Correct Answer)

Open solution

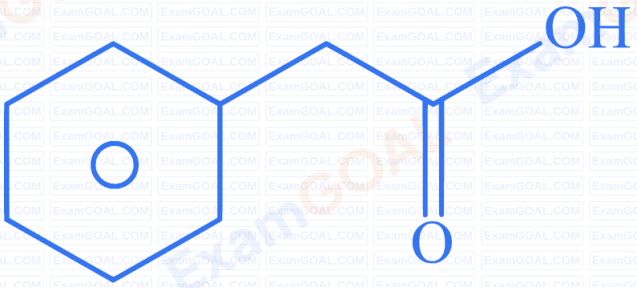
The final product A formed in the following multistep reaction sequence is



(A)



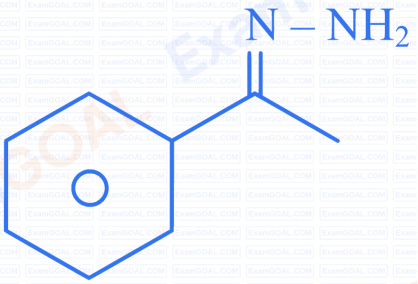
(B)



(C)



(D)



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Q14. MCQ (Single Correct Answer)

Which of the following is not correct?

(A) ΔG is positive for a spontaneous reaction

(B) ΔG is positive for a non-spontaneous reaction

(C) ΔG is zero for a reversible reaction

(D) ΔG is negative for a spontaneous reaction

Open solution

Q15. MCQ (Single Correct Answer)

The interaction between π bond and lone pair of electrons present on an adjacent atom is responsible for

(A) Resonance effect

(B) Electromeric effect

(C) Hyperconjugation

(D) Inductive effect

Open solution

Q16. MCQ (Single Correct Answer)

In which one of the following metal carbonyls, CO forms a bridge between metal atoms?

(A) $[Os_3(CO)_{12}]$

(B) $[Ru_3(CO)_{12}]$

(C) $[Mn_2(CO)_{10}]$

(D) $[Co_2(CO)_8]$

Open solution

Q17. MCQ (Single Correct Answer)

$KMnO_4$ decomposes on heating at $513\ K$ to form O_2 along with

(A) K_2MnO_4 & Mn

(B) MnO_2 & K_2O_2

(C) K_2MnO_4 & MnO_2

(D) Mn & KO_2

Open solution

Q18. MCQ (Single Correct Answer)

Appearance of blood red colour, on treatment of the sodium fusion extract of an organic compound with $FeSO_4$ in presence of concentrated H_2SO_4 indicates the presence of element/s

(A) Br

(B) S

(C) N and S

(D) N

Open solution

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Q19. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R :

Assertion A : Aryl halides cannot be prepared by replacement of hydroxyl group of phenol by halogen atom.

Reason R : Phenols react with halogen acids violently.

In the light of the above statements, choose the **most appropriate** from the options given below :

(A) A is true but R is false

(B) Both A and R are true but R is NOT the correct explanation of A

(C) A is false but R is true

(D) Both A and R are true and R is the correct explanation of A

Q20. MCQ (Single Correct Answer)

Open solution

In alkaline medium, MnO_4^- oxidises I^- to

(A) I_2 (B) IO_3^- (C) IO^- (D) IO_4^-

Q21. Numerical

Open solution

The mass of zinc produced by the electrolysis of zinc sulphate solution with a steady current of $0.015\ A$ for 15 minutes is _____ $\times 10^{-4}$ g.

(Atomic mass of zinc $= 65.4$ amu)

Q22. Numerical

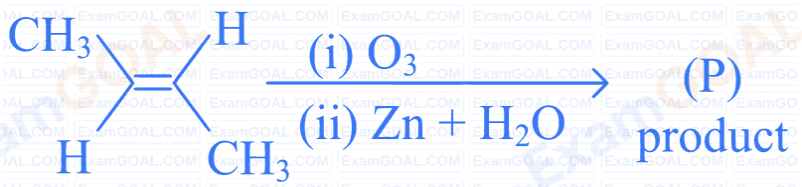
Open solution

A solution of H_2SO_4 is $31.4\% H_2SO_4$ by mass and has a density of $1.25\ g/mL$. The molarity of the H_2SO_4 solution is _____ M (nearest integer)

[Given molar mass of $H_2SO_4 = 98\ g\ mol^{-1}$]

Q23. Numerical

Open solution



Consider the given reaction. The total number of oxygen atom/s present per molecule of the product (P) is _____.

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Q24. Numerical

For the reaction $N_2O_{4(g)} \rightleftarrows 2NO_{2(g)}$, $K_p = 0.492\ atm$ at $300\ K$. K_c for the reaction at same temperature is _____ $\times 10^{-2}$.

(Given : $R = 0.082\ L\ atm\ mol^{-1}\ K^{-1}$)

Open solution

Q25. Numerical

The number of species from the following which are paramagnetic and with bond order equal to one is _____.

$H_2, He_2^+, O_2^+, N_2^{2-}, O_2^{2-}, F_2, Ne_2^+, B_2$

Open solution

Q26. Numerical

For a reaction taking place in three steps at same temperature, overall rate constant $K = \frac{K_1 K_2}{K_3}$. If Ea_1, Ea_2 and Ea_3 are 40, 50 and $60\ kJ/mol$ respectively, the overall Ea is _____ kJ/mol.

Open solution

Q27. Numerical

From the compounds given below, number of compounds which give positive Fehling's test is _____.

Benzaldehyde, Acetaldehyde, Acetone, Acetophenone, Methanal, 4-nitrobenzaldehyde, cyclohexane carbaldehyde.

Open solution

Q28. Numerical

Number of compounds with one lone pair of electrons on central atom amongst following is _____.

$O_3, H_2O, SF_4, ClF_3, NH_3, BrF_5, XeF_4$

Open solution

Q29. Numerical

Number of compounds among the following which contain sulphur as heteroatom is _____.

Furan, Thiophene, Pyridine, Pyrrole, Cysteine, Tyrosine

Open solution

Q30. Numerical

The osmotic pressure of a dilute solution is $7 \times 10^5\ Pa$ at $273\ K$. Osmotic pressure of the same solution at $283\ K$ is _____ $\times 10^4 Nm^{-2}$.

Open solution

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Mathematics 30 questions

Q31. MCQ (Single Correct Answer)

Suppose $f(x)=\frac{(2^x+2^{-x})\tan x\sqrt{\tan^{-1}(x^2-x+1)}}{(7x^2+3x+1)^3}$. Then the value of $f'(0)$ is equal to

(A) π (B) $\sqrt{\pi}$ (C) 0 (D) $\frac{\pi}{2}$

Open solution

Q32. MCQ (Single Correct Answer)

Let $\vec{a},\vec{b}$ and $\vec{c}$ be three non-zero vectors such that $\vec{b}$ and $\vec{c}$ are non-collinear. If $\vec{a}+5\vec{b}$ is collinear with $\vec{c},\vec{b}+6\vec{c}$ is collinear with $\vec{a}$ and $\vec{a}+\alpha\vec{b}+\beta\vec{c}=\overrightarrow{0}$, then $\alpha+\beta$ is equal to

(A) 30 (B) -30 (C) -25 (D) 35

Open solution

Q33. MCQ (Single Correct Answer)

Let $\left(5,\frac{a}{4}\right)$ be the circumcenter of a triangle with vertices $\mathrm{A}(a,-2),\mathrm{B}(a,6)$ and $C\left(\frac{a}{4},-2\right)$. Let α denote the circumradius, β denote the area and γ denote the perimeter of the triangle. Then $\alpha+\beta+\gamma$ is

(A) 60 (B) 62 (C) 53 (D) 30

Open solution

Q34. MCQ (Single Correct Answer)

If $\alpha,-\frac{\pi}{2}<\alpha<\frac{\pi}{2}$ is the solution of $4\cos\theta+5\sin\theta=1$, then the value of $\tan\alpha$ is

(A) $\frac{10-\sqrt{10}}{12}$ (B) $\frac{\sqrt{10}-10}{6}$ (C) $\frac{\sqrt{10}-10}{12}$ (D) $\frac{10-\sqrt{10}}{6}$

Open solution

Q35. MCQ (Single Correct Answer)

If in a G.P. of 64 terms, the sum of all the terms is 7 times the sum of the odd terms of the G.P, then the common ratio of the G.P. is equal to

(A) 7 (B) 6 (C) 5 (D) 4

Open solution

Q36. MCQ (Single Correct Answer)

A fair die is thrown until 2 appears. Then the probability, that 2 appears in even number of throws, is

(A) $\frac{5}{11}$ (B) $\frac{5}{6}$ (C) $\frac{1}{6}$ (D) $\frac{6}{11}$

Open solution

Q37. MCQ (Single Correct Answer)

In an A.P., the sixth term $a_6=2$. If the product $a_1a_4a_5$ is the greatest, then the common difference of the A.P. is equal to

(A) $\frac{2}{3}$ (B) $\frac{5}{8}$ (C) $\frac{3}{2}$ (D) $\frac{8}{5}$

Open solution

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Q38. MCQ (Single Correct Answer)

Let $A=\begin{bmatrix}1 & 0 & 0\\ 0 & \alpha & \beta\\ 0 & \beta & \alpha\end{bmatrix}$ and $|2\,\mathrm{A}|^3=2^{21}$ where $\alpha,\beta\in Z$. Then a value of α is

(A) 9 (B) 17 (C) 3 (D) 5

Open solution

Q39. MCQ (Single Correct Answer)

Let R be a relation on $Z\times Z$ defined by $(a,b)R(c,d)$ if and only if $ad-bc$ is divisible by 5. Then R is

(A) Reflexive and transitive but not symmetric

(B) Reflexive and symmetric but not transitive

(C) Reflexive but neither symmetric nor transitive

(D) Reflexive, symmetric and transitive

Open solution

Q40. MCQ (Single Correct Answer)

If $f(x)=\begin{cases}2+2x, & -1\le x<0\\ 1-\frac{x}{3}, & 0\le x\le 3\end{cases}; g(x)=\begin{cases}-x, & -3\le x\le 0\\ x, & 0<x\le 1\end{cases}$, then range of $(fog)(x)$ is

(A) $[0,1)$ (B) $[0,3)$ (C) $(0,1]$ (D) $[0,1]$

Open solution

Q41. MCQ (Single Correct Answer)

Let O be the origin and the position vectors of A and B be $2\hat{i}+2\hat{j}+\hat{k}$ and $2\hat{i}+4\hat{j}+4\hat{k}$ respectively. If the internal bisector of $\angle\mathrm{AOB}$ meets the line AB at C, then the length of OC is

(A) $\frac{3}{2}\sqrt{34}$ (B) $\frac{2}{3}\sqrt{31}$ (C) $\frac{2}{3}\sqrt{34}$ (D) $\frac{3}{2}\sqrt{31}$

Open solution

Q42. MCQ (Single Correct Answer)

In a $\triangle ABC$, suppose $y=x$ is the equation of the bisector of the angle B and the equation of the side AC is $2x-y=2$. If $2AB=BC$ and the points A and B are respectively $(4,6)$ and (α,β), then $\alpha+2\beta$ is equal to

(A) 42 (B) 39 (C) 48 (D) 45

Open solution

Q43. MCQ (Single Correct Answer)

For $x\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$, if $y(x)=\int\frac{\operatorname{cosec}x+\sin x}{\operatorname{cosec}x\sec x+\tan x\sin^2 x}dx$, and $\lim\limits_{x\to\left(\frac{\pi}{2}\right)^-}y(x)=0$ then $y\left(\frac{\pi}{4}\right)$ is equal to

(A) $-\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (B) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$

(C) $\frac{1}{2}\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (D) $\frac{1}{\sqrt{2}}\tan^{-1}\left(-\frac{1}{2}\right)$

Open solution

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Q44. MCQ (Single Correct Answer)

Let A be a square matrix such that $\mathrm{AA}^{\mathrm{T}}=\mathrm{I}$. Then $\frac{1}{2}A\left[\left(A+A^{T}\right)^{2}+\left(A-A^{T}\right)^{2}\right]$ is equal to

(A) $\mathrm{A}^{2}+\mathrm{A}^{\mathrm{T}}$ (B) $\mathrm{A}^{3}+\mathrm{I}$

(C) $\mathrm{A}^{3}+\mathrm{A}^{\mathrm{T}}$ (D) $\mathrm{A}^{2}+\mathrm{I}$

Open solution

Q45. MCQ (Single Correct Answer)

If $z=\frac{1}{2}-2i$ is such that $|z+1|=\alpha z+\beta(1+i), i=\sqrt{-1}$ and $\alpha, \beta \in \mathbb{R}$, then $\alpha+\beta$ is equal to

(A) 2 (B) -4 (C) 3 (D) -1

Open solution

Q46. MCQ (Single Correct Answer)

Consider the function $f:\left[\frac{1}{2}, 1\right] \rightarrow \mathbb{R}$ defined by $f(x)=4\sqrt{2}x^{3}-3\sqrt{2}x-1$. Consider the statements

(I) The curve $y=f(x)$ intersects the x-axis exactly at one point.

(II) The curve $y=f(x)$ intersects the x-axis at $x=\cos\frac{\pi}{12}$.

Then

(A) Both (I) and (II) are correct. (B) Only (I) is correct.

(C) Both (I) and (II) are incorrect. (D) Only (II) is correct.

Open solution

Q47. MCQ (Single Correct Answer)

Let PQR be a triangle with $R(-1,4,2)$. Suppose $M(2,1,2)$ is the mid point of PQ. The distance of the centroid of $\triangle$PQR from the point of intersection of the lines $\frac{x-2}{0}=\frac{y}{2}=\frac{z+3}{-1}$ and $\frac{x-1}{1}=\frac{y+3}{-3}=\frac{z+1}{1}$ is

(A) 69 (B) $\sqrt{99}$ (C) $\sqrt{69}$ (D) 9

Open solution

Q48. MCQ (Single Correct Answer)

$\lim\limits_{x\to\frac{\pi}{2}}\left(\frac{1}{\left(x-\frac{\pi}{2}\right)^{2}}\int\limits_{x^{3}}^{\left(\frac{\pi}{2}\right)^{3}}\cos\left(t^{\frac{1}{3}}\right)dt\right)$ is equal to

(A) $\frac{3\pi^{2}}{4}$ (B) $\frac{3\pi^{2}}{8}$ (C) $\frac{3\pi}{4}$ (D) $\frac{3\pi}{8}$

Open solution

Q49. MCQ (Single Correct Answer)

A function $y=f(x)$ satisfies $f(x)\sin 2x+\sin x-\left(1+\cos^{2}x\right)f'(x)=0$ with condition $f(0)=0$. Then, $f\left(\frac{\pi}{2}\right)$ is equal to

(A) 2 (B) 1 (C) -1 (D) 0

Open solution

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Q50. MCQ (Single Correct Answer)

If the value of the integral $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{x^2\cos x}{1+\pi^x} + \frac{1+\sin^2 x}{1+e^{\sin x^{2023}}}\right)dx = \frac{\pi}{4}(\pi+a) - 2$, then the value of a is

(A) $-\frac{3}{2}$ (B) 3 (C) $\frac{3}{2}$ (D) 2

Open solution

Q51. Numerical

The area (in sq. units) of the part of the circle $x^2+y^2=169$ which is below the line $5x-y=13$ is $\frac{\pi\alpha}{2\beta} - \frac{65}{2} + \frac{\alpha}{\beta}\sin^{-1}\left(\frac{12}{13}\right)$, where α, β are coprime numbers. Then $\alpha+\beta$ is equal to _____.

Open solution

Q52. Numerical

If the mean and variance of the data $65, 68, 58, 44, 48, 45, 60, \alpha, \beta, 60$ where $\alpha > \beta$, are 56 and 66.2 respectively, then $\alpha^2+\beta^2$ is equal to _____.

Open solution

Q53. Numerical

Equations of two diameters of a circle are $2x-3y=5$ and $3x-4y=7$. The line joining the points $\left(-\frac{22}{7}, -4\right)$ and $\left(-\frac{1}{7}, 3\right)$ intersects the circle at only one point $P(\alpha, \beta)$. Then, $17\beta-\alpha$ is equal to _____.

Open solution

Q54. Numerical

A line with direction ratios $2, 1, 2$ meets the lines $x=y+2=z$ and $x+2=2y=2z$ respectively at the points P and Q. If the length of the perpendicular from the point $(1, 2, 12)$ to the line PQ is l, then l^2 is _____.

Open solution

Q55. Numerical

If $\frac{{}^{11}C_1}{2} + \frac{{}^{11}C_2}{3} + \ldots + \frac{{}^{11}C_9}{10} = \frac{n}{m}$ with $\gcd(n, m) = 1$, then $n+m$ is equal to _____.

Open solution

Q56. Numerical

If the points of intersection of two distinct conics $x^2+y^2=4b$ and $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ lie on the curve $y^2=3x^2$, then $3\sqrt{3}$ times the area of the rectangle formed by the intersection points is _____.

Open solution

Q57. Numerical

All the letters of the word "GTWENTY" are written in all possible ways with or without meaning and these words are written as in a dictionary. The serial number of the word "GTWENTY" is _____.

Open solution

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Q58. Numerical

Let α, β be the roots of the equation $x^2 - x + 2 = 0$ with $\mathrm{Im}(\alpha) > \mathrm{Im}(\beta)$. Then $\alpha^6 + \alpha^4 + \beta^4 - 5\alpha^2$ is equal to _____.

Open solution

Q59. Numerical

Let $f(x) = 2^x - x^2, x \in \mathbb{R}$. If m and n are respectively the number of points at which the curves $y = f(x)$ and $y = f'(x)$ intersect the x-axis, then the value of $\mathrm{m} + \mathrm{n}$ is _____.

Open solution

Q60. Numerical

If the solution curve $y = y(x)$ of the differential equation $\left(1 + y^2\right)\left(1 + \log_e x\right)dx + xdy = 0, x > 0$ passes through the point $(1, 1)$ and $y(e) = \frac{\alpha - \tan\left(\frac{3}{2}\right)}{\beta + \tan\left(\frac{3}{2}\right)}$, then $\alpha + 2\beta$ is _____.

Open solution

Physics 30 questions

Q61. MCQ (Single Correct Answer)

The resistance $R = \frac{V}{I}$ where $\mathrm{V} = (200 \pm 5)\mathrm{V}$ and $I = (20 \pm 0.2)\mathrm{A}$, the percentage error in the measurement of R is :

(A) 5.5% (B) 3% (C) 7% (D) 3.5%

Open solution

Q62. MCQ (Single Correct Answer)

The de-Broglie wavelength of an electron is the same as that of a photon. If velocity of electron is 25% of the velocity of light, then the ratio of K.E. of electron and K.E. of photon will be:

(A) $\frac{1}{4}$ (B) $\frac{8}{1}$ (C) $\frac{1}{8}$ (D) $\frac{1}{1}$

Open solution

Q63. MCQ (Single Correct Answer)

A convex mirror of radius of curvature $30\ \mathrm{cm}$ forms an image that is half the size of the object. The object distance is :

(A) -45 cm (B) -15 cm (C) 45 cm (D) 15 cm

Open solution

Q64. MCQ (Single Correct Answer)

The deflection in moving coil galvanometer falls from 25 divisions to 5 divisions when a shunt of 24Ω is applied. The resistance of galvanometer coil will be :

(A) 48Ω (B) 100Ω (C) 96Ω (D) 12Ω

Open solution

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Q65. MCQ (Single Correct Answer)

A capacitor of capacitance $100\mu\text{F}$ is charged to a potential of 12 V and connected to a 6.4 mH inductor to produce oscillations. The maximum current in the circuit would be :

(A) 2.0 A (B) 3.2 A (C) 1.5 A (D) 1.2 A

Open solution

Q66. MCQ (Single Correct Answer)

At what distance above and below the surface of the earth a body will have same weight. (take radius of earth as R.)

(A) $\frac{\sqrt{3}R-R}{2}$ (B) $\frac{R}{2}$ (C) $\frac{\sqrt{5}R-R}{2}$ (D) $\sqrt{5}R-R$

Open solution

Q67. MCQ (Single Correct Answer)

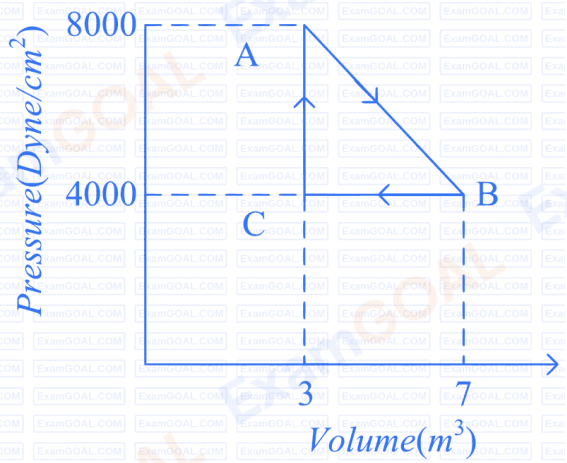
A galvanometer having coil resistance 10Ω shows a full scale deflection for a current of 3 mA. For it to measure a current of 8 A, the value of the shunt should be:

(A) $3.75\times10^{-3}\Omega$ (B) $3\times10^{-3}\Omega$ (C) $4.85\times10^{-3}\Omega$ (D) $2.75\times10^{-3}\Omega$

Open solution

Q68. MCQ (Single Correct Answer)

A thermodynamic system is taken from an original state A to an intermediate state B by a linear process as shown in the figure. Its volume is then reduced to the original value from B to C by an isobaric process. The total work done by the gas from A to B and B to C would be :



(A) 800 J (B) 2200 J (C) 33800 J (D) 1200 J

Open solution

Q69. MCQ (Single Correct Answer)

Two charges of $5Q$ and $-2Q$ are situated at the points $(3a,0)$ and $(-5a,0)$ respectively. The electric flux through a sphere of radius '$4a$' having center at origin is :

(A) $\frac{2Q}{\varepsilon_0}$ (B) $\frac{7Q}{\varepsilon_0}$ (C) $\frac{3Q}{\varepsilon_0}$ (D) $\frac{5Q}{\varepsilon_0}$

Open solution

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Q70. MCQ (Single Correct Answer)

A biconvex lens of refractive index 1.5 has a focal length of $20\ \text{cm}$ in air. Its focal length when immersed in a liquid of refractive index 1.6 will be:

(A) -16 cm (B) +16 cm (C) +160 cm (D) -160 cm

Open solution

Q71. MCQ (Single Correct Answer)

A block of mass $100\ \text{kg}$ slides over a distance of $10\ \text{m}$ on a horizontal surface. If the co-efficient of friction between the surfaces is 0.4, then the work done against friction $(\text{in } J)$ is :

(A) 3900 (B) 4500 (C) 4200 (D) 4000

Open solution

Q72. MCQ (Single Correct Answer)

Given below are two statements:

Statement I : If a capillary tube is immersed first in cold water and then in hot water, the height of capillary rise will be smaller in hot water.

Statement II : If a capillary tube is immersed first in cold water and then in hot water, the height of capillary rise will be smaller in cold water.

In the light of the above statements, choose the most appropriate from the options given below

(A) Both Statement I and Statement II are false

(B) Both Statement I and Statement II are true

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Open solution

Q73. MCQ (Single Correct Answer)

If the radius of curvature of the path of two particles of same mass are in the ratio $3:4$, then in order to have constant centripetal force, their velocities will be in the ratio of :

(A) $1:\sqrt{3}$ (B) $2:\sqrt{3}$ (C) $\sqrt{3}:2$ (D) $\sqrt{3}:1$

Open solution

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Q74. MCQ (Single Correct Answer)

Open solution

Match List I with List II

	List - I		List - II
(A)	$\oint \vec{B} \cdot \overrightarrow{dl} = \mu_o i_c + \mu_o \varepsilon_o \frac{d\phi_E}{dt}$	(I)	Gauss' law for electricity
(B)	$\oint \vec{E} \cdot \overrightarrow{dl} = \frac{d\phi_B}{dt}$	(II)	Gauss' law for magnetism
(C)	$\oint \vec{E} \cdot \overrightarrow{dA} = \frac{Q}{\varepsilon_o}$	(III)	Faraday law
(D)	$\oint \vec{B} \cdot \overrightarrow{dA} = 0$	(IV)	Ampere - Maxwell law

Choose the correct answer from the options given below:

(A) A-IV, B-III, C-I, D-II

(B) A-I, B-II, C-III, D-IV

(C) A-IV, B-I, C-III, D-II

(D) A-II, B-III, C-I, D-IV

Q75. MCQ (Single Correct Answer)

Open solution

Two vessels A and B are of the same size and are at same temperature. A contains 1 g of hydrogen and B contains 1 g of oxygen. $\mathrm{P_A}$ and $\mathrm{P_B}$ are the pressures of the gases in A and B respectively, then $\frac{P_A}{P_B}$ is:

(A) 4

(B) 32

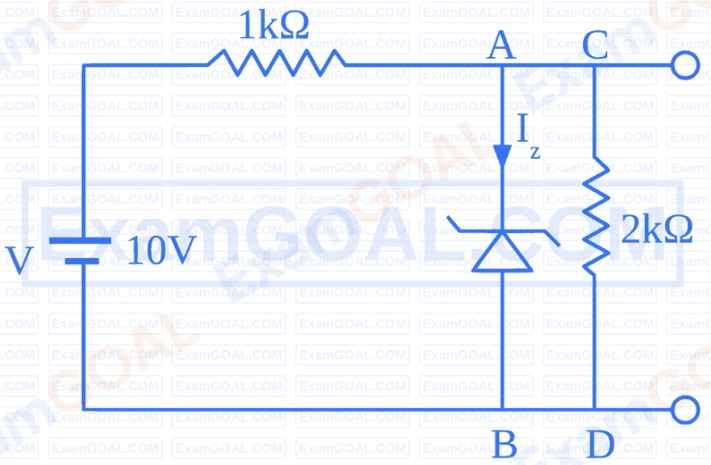
(C) 8

(D) 16

Q76. MCQ (Single Correct Answer)

Open solution

In the given circuit, the breakdown voltage of the Zener diode is 3.0 V. What is the value of $\mathrm{I_z}$?



(A) 3.3 mA

(B) 10 mA

(C) 5.5 mA

(D) 7 mA

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Q77. MCQ (Single Correct Answer)

A body starts moving from rest with constant acceleration covers displacement S_1 in first $(p-1)$ seconds and S_2 in first p seconds. The displacement $\mathrm{S}_1+\mathrm{S}_2$ will be made in time :

Open solution

(A) $(2p+1)s$

(B) $(2p-1)s$

(C) $\left(2p^2-2p+1\right)s$

(D) $\sqrt{(2p^2-2p+1)}s$

Q78. MCQ (Single Correct Answer)

The electric current through a wire varies with time as $I=I_0+\beta t$, where $I_0=20\ \mathrm{A}$ and $\beta=3\ \mathrm{A/s}$. The amount of electric charge crossed through a section of the wire in $20\ \mathrm{s}$ is :

Open solution

(A) 80 C

(B) 800 C

(C) 1000 C

(D) 1600 C

Q79. MCQ (Single Correct Answer)

The potential energy function (in J) of a particle in a region of space is given as $U=\left(2x^2+3y^3+2z\right)$. Here x, y and z are in meter. The magnitude of x-component of force (in N) acting on the particle at point $P(1,2,3)\mathrm{m}$ is :

Open solution

(A) 4

(B) 2

(C) 8

(D) 6

Q80. MCQ (Single Correct Answer)

The explosive in a Hydrogen bomb is a mixture of ${}_1\mathrm{H}^2$, ${}_1\mathrm{H}^3$ and ${}_3\mathrm{Li}^6$ in some condensed form. The chain reaction is given by

Open solution

${}_3\mathrm{Li}^6+{}_0\mathrm{n}^1 \rightarrow {}_2\mathrm{He}^4+{}_1\mathrm{H}^3$

${}_1\mathrm{H}^2+{}_1\mathrm{H}^3 \rightarrow {}_2\mathrm{He}^4+{}_0\mathrm{n}^1$

During the explosion the energy released is approximately

[Given ; $\mathrm{M(Li)}=6.01690\ \mathrm{amu}, \mathrm{M}\left({}_1\mathrm{H}^2\right)=2.01471\ \mathrm{amu}, \mathrm{M}\left({}_2\mathrm{He}^4\right)=4.00388\ \mathrm{amu}$, and $1\ \mathrm{amu}=931.5\ \mathrm{MeV}$]

(A) 22.22 MeV

(B) 28.12 MeV

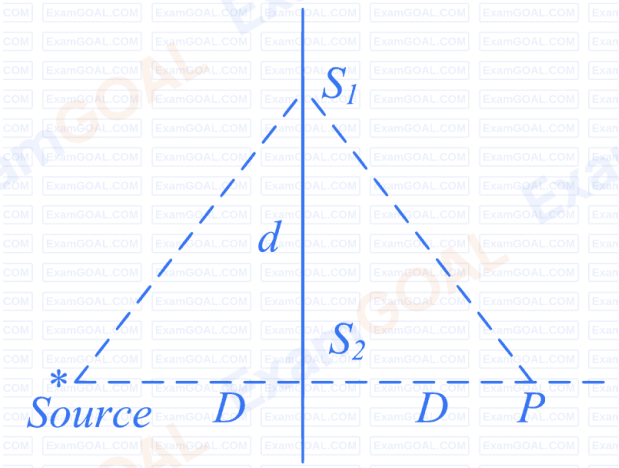
(C) 16.48 MeV

(D) 12.64 MeV

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Q81. Numerical

In a double slit experiment shown in figure, when light of wavelength $400\ \text{nm}$ is used, dark fringe is observed at P. If $\text{D} = 0.2\ \text{m}$, the minimum distance between the slits S_1 and S_2 is _____ mm.



Open solution

Q82. Numerical

A cylinder is rolling down on an inclined plane of inclination 60°. Its acceleration during rolling down will be $\frac{x}{\sqrt{3}} m/s^2$, where $x =$ _____ (use $\text{g} = 10\ \text{m/s}^2$).

Open solution

Q83. Numerical

An electron is moving under the influence of the electric field of a uniformly charged infinite plane sheet S having surface charge density $+\sigma$. The electron at $t = 0$ is at a distance of $1\ \text{m}$ from S and has a speed of $1\ \text{m/s}$. The maximum value of σ if the electron strikes S at $t = 1\ \text{s}$ is $\alpha \left[\frac{m\epsilon_0}{e}\right] \frac{C}{m^2}$, the value of α is _____.

Open solution

Q84. Numerical

When a hydrogen atom going from $n = 2$ to $n = 1$ emits a photon, its recoil speed is $\frac{x}{5}\ \text{m/s}$. Where $x =$ _____. (Use, mass of hydrogen atom $= 1.6 \times 10^{-27}\ \text{kg}$)

Open solution

Q85. Numerical

A square loop of side $10\ \text{cm}$ and resistance 0.7Ω is placed vertically in east-west plane. A uniform magnetic field of $0.20T$ is set up across the plane in north east direction. The magnetic field is decreased to zero in $1\ \text{s}$ at a steady rate. Then, magnitude of induced emf is $\sqrt{x} \times 10^{-3}\ \text{V}$. The value of x is _____.

Open solution

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Q86. Numerical

The magnetic potential due to a magnetic dipole at a point on its axis situated at a distance of $20\ \text{cm}$ from its center is $1.5 \times 10^{-5}\ \text{T m}$. The magnetic moment of the dipole is _____ $A\,\text{m}^2$.
(Given : $\frac{\mu_o}{4\pi} = 10^{-7}\text{Tm}A^{-1}$)

Open solution

Q87. Numerical

A 16Ω wire is bent to form a square loop. A $9\ \text{V}$ battery with internal resistance 1Ω is connected across one of its sides. If a $4\mu F$ capacitor is connected across one of its diagonals, the energy stored by the capacitor will be $\frac{x}{2}\mu J$, where $x =$ _____

Open solution

Q88. Numerical

When the displacement of a simple harmonic oscillator is one third of its amplitude, the ratio of total energy to the kinetic energy is $\frac{x}{8}$, where $x =$ _____.

Open solution

Q89. Numerical

A ball rolls off the top of a stairway with horizontal velocity u. The steps are $0.1\ \text{m}$ high and $0.1\ \text{m}$ wide. The minimum velocity u with which that ball just hits the step 5 of the stairway will be $\sqrt{x}\ \text{ms}^{-1}$ where $x =$ _____ [use $\text{g} = 10\ \text{m/s}^2$].

Open solution

Q90. Numerical

In a test experiment on a model aeroplane in wind tunnel, the flow speeds on the upper and lower surfaces of the wings are $70\ \text{ms}^{-1}$ and $65\ \text{ms}^{-1}$ respectively. If the wing area is $2\ \text{m}^2$, the lift of the wing is _____ N.
(Given density of air $= 1.2\ \text{kg m}^{-3}$)

Open solution

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Answer Key

Chemistry

1. **D**	2. **A**	3. **C**	4. **A**	5. **D**
6. **A**	7. **B**	8. **A**	9. **C**	10. **B**
11. **B**	12. **A**	13. **A**	14. **A**	15. **A**
16. **D**	17. **C**	18. **C**	19. **A**	20. **B**
21. **46**	22. **4**	23. **1**	24. **2**	25. **1**
26. **30**	27. **3**	28. **4**	29. **2**	30. **73**

Mathematics

31. **B**	32. **D**	33. **C**	34. **C**	35. **B**
36. **A**	37. **D**	38. **D**	39. **B**	40. **D**
41. **C**	42. **A**	43. **D**	44. **C**	45. **C**
46. **A**	47. **C**	48. **B**	49. **B**	50. **B**
51. **171**	52. **6344**	53. **2**	54. **65**	55. **2041**
56. **432**	57. **553**	58. **13**	59. **5**	60. **3**

Physics

61. **D**	62. **C**	63. **B**	64. **C**	65. **C**
66. **C**	67. **A**	68. **A**	69. **D**	70. **D**
71. **D**	72. **C**	73. **C**	74. **A**	75. **D**
76. **C**	77. **D**	78. **C**	79. **A**	80. **A**
81. **0.20**	82. **10**	83. **8**	84. **17**	85. **2**
86. **6**	87. **81**	88. **9**	89. **2**	90. **810**

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